Data Visualization Lab

Experiment No. : 2

Create Basic Visualizations in Tableau/PowerBI

Experiment No.2

1. **Aim:** Create bar charts, line graphs, and pie charts using a sample dataset in Tableau/PowerBI applying effective visualization principles.

2. **Objectives:**

   - Develop skills to both design and critique visualizations.
   - Understand why visualization is an important part of data analysis.

3. **Course Outcomes:** Students will be able to understand the fundamentals of data visualization and its importance.

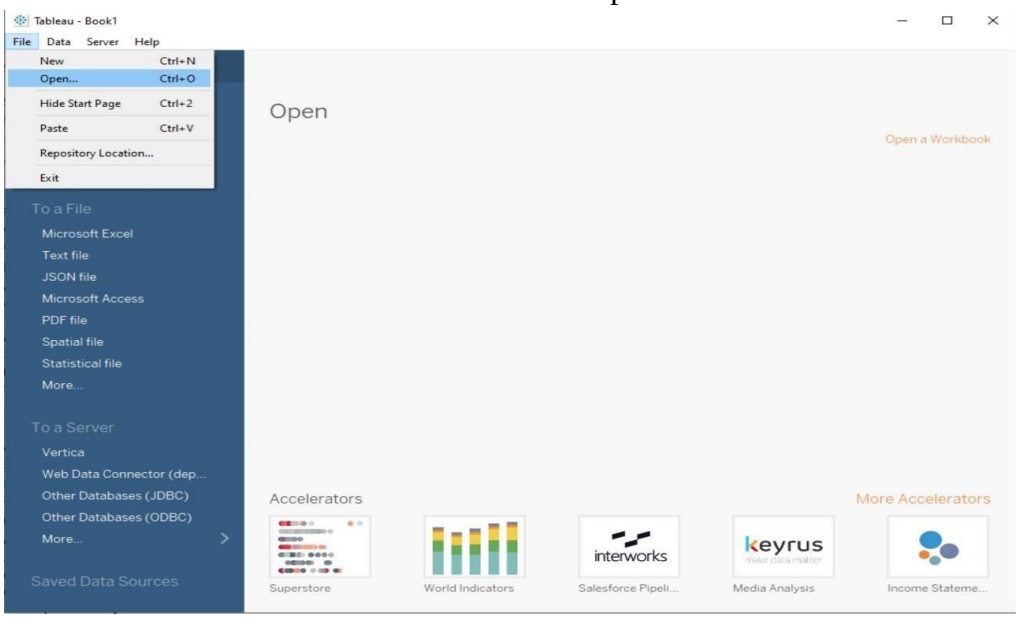
4. **Hardware / Software Required: Tableau/PowerBI**

5. **Theory:**

   Data Visualization Charts is a method of presenting information and data in a visual way using graphical representation elements like charts, graphs and maps. These visual elements help users easily understand complex datasets quickly and efficiently. There are many different types of visualization charts, each best suited for different data types and analysis of data provided.
   The basic visualizations include: Bar Chart, Line Chart, Pie Chart and Scatter Plot.

- **Implementation on Tableau:**

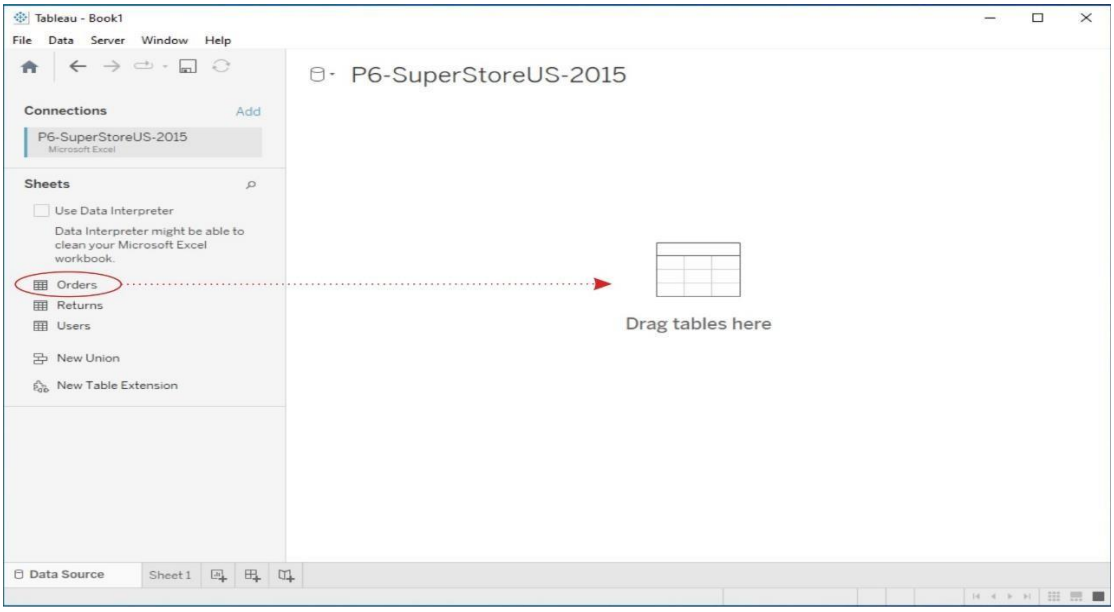
1. Connect Your Data to Tableau:
   - Launch Tableau Desktop.
   - Go to "File" Menu and then click on "Open".



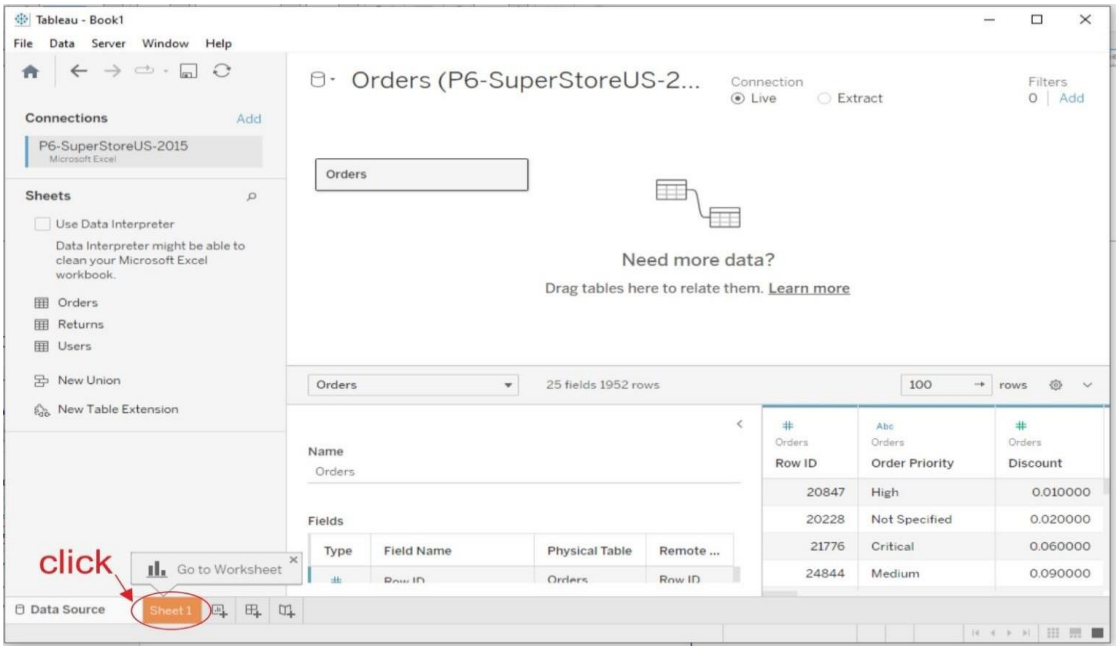
1.1 Choose the data source type (e.g., Excel, CSV, text file) and Select the data

file(P6-SuperStoreUS-2015.xls) and click "Open".

1.2 Drag any table(e.g. Orders) into working area.

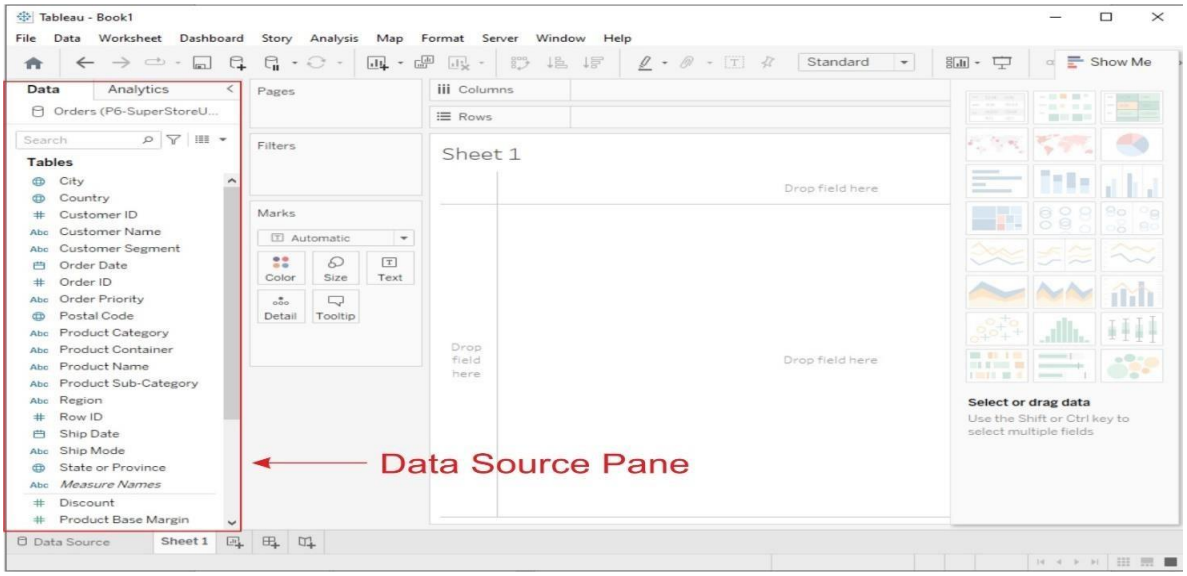


1.3 Click on Worksheet(**Sheet1**).



2. Data Source Pane:

Once your data is connected, the Data Source Pane will appear on the left-hand side of the Tableau interface. Here, you can see a preview of your data and perform data transformations or join multiple data sources if necessary.

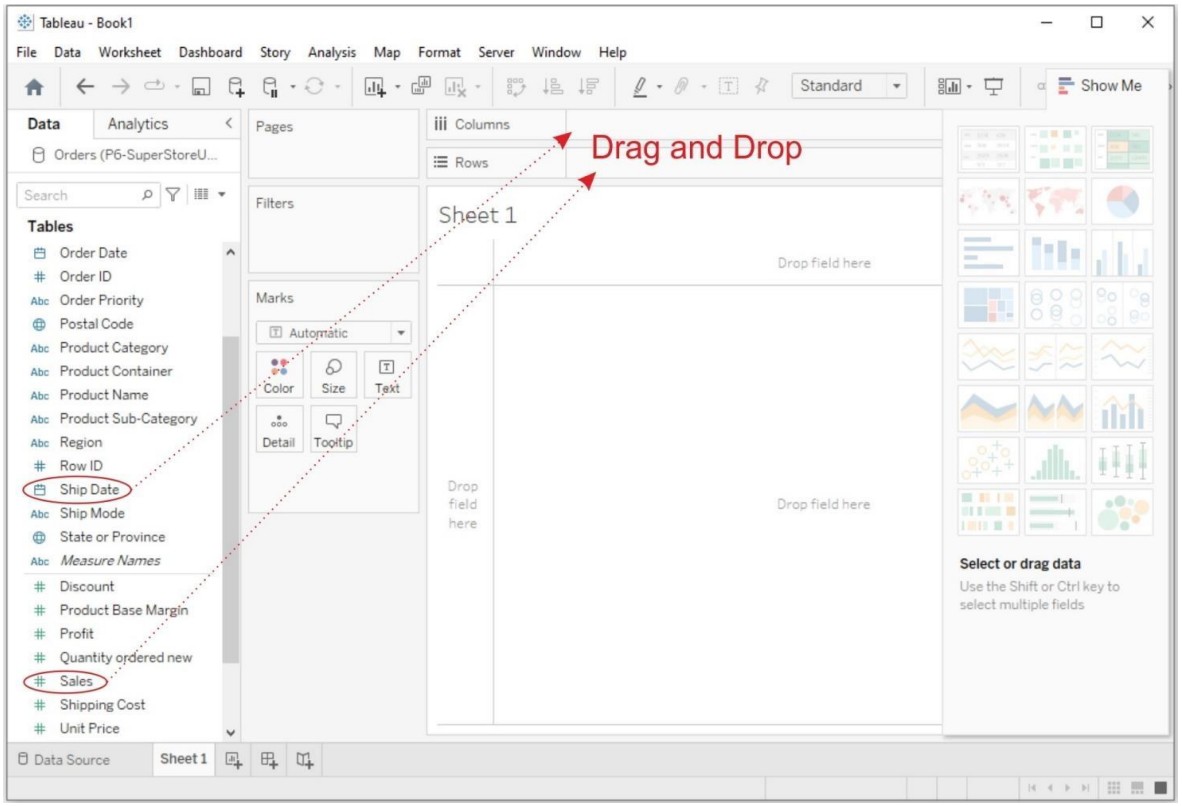


3. Creating Basic Charts:

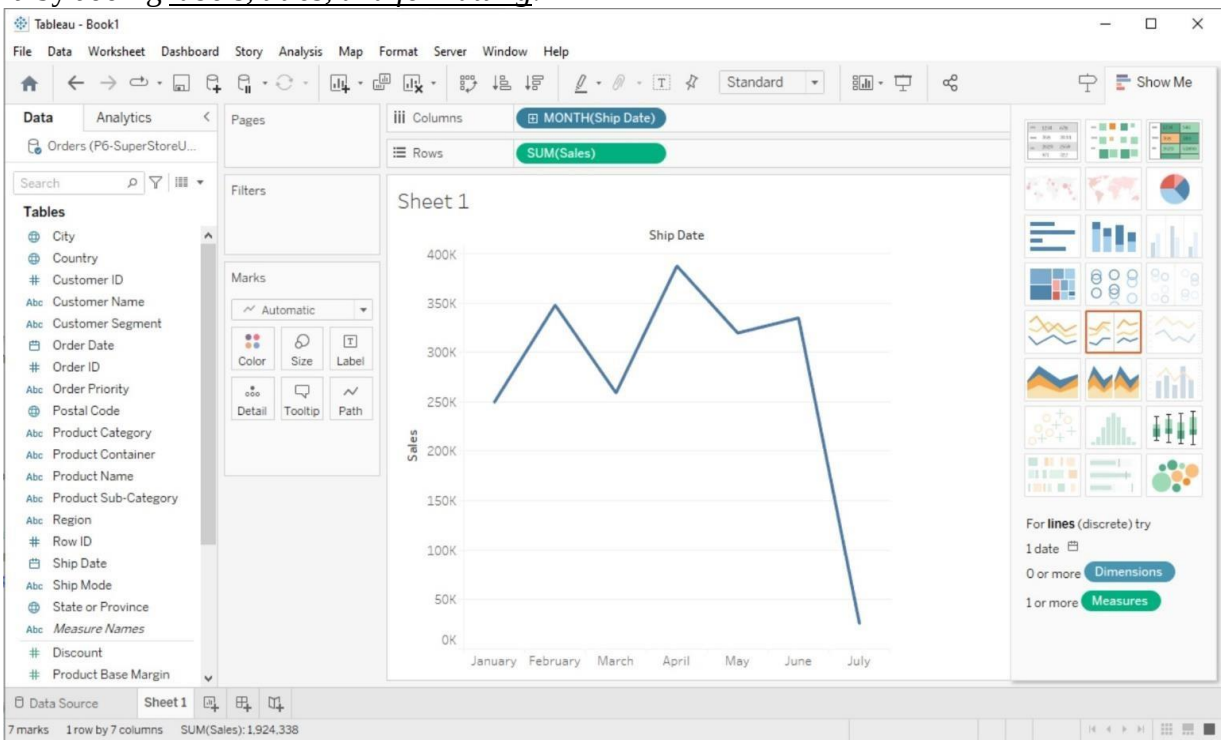
Now, let's create some basic charts using Tableau:

a. Line Chart:

1. From the "**Data Source pane**", drag and drop the date field to the **Columns shelf** and a numeric field (e.g., sales, revenue) to the **Rows shelf**.

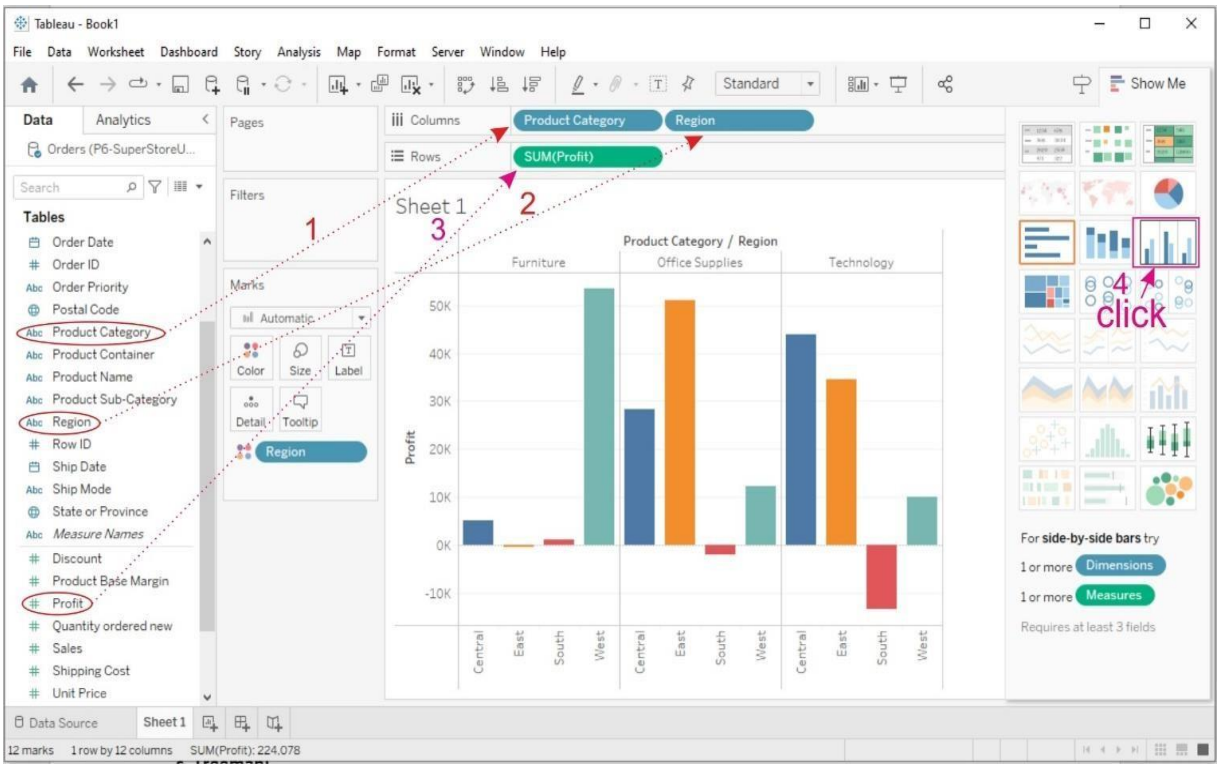


2. Then **Tableau** will automatically create a line chart. You can customize it by adding *labels, titles, and formatting*.

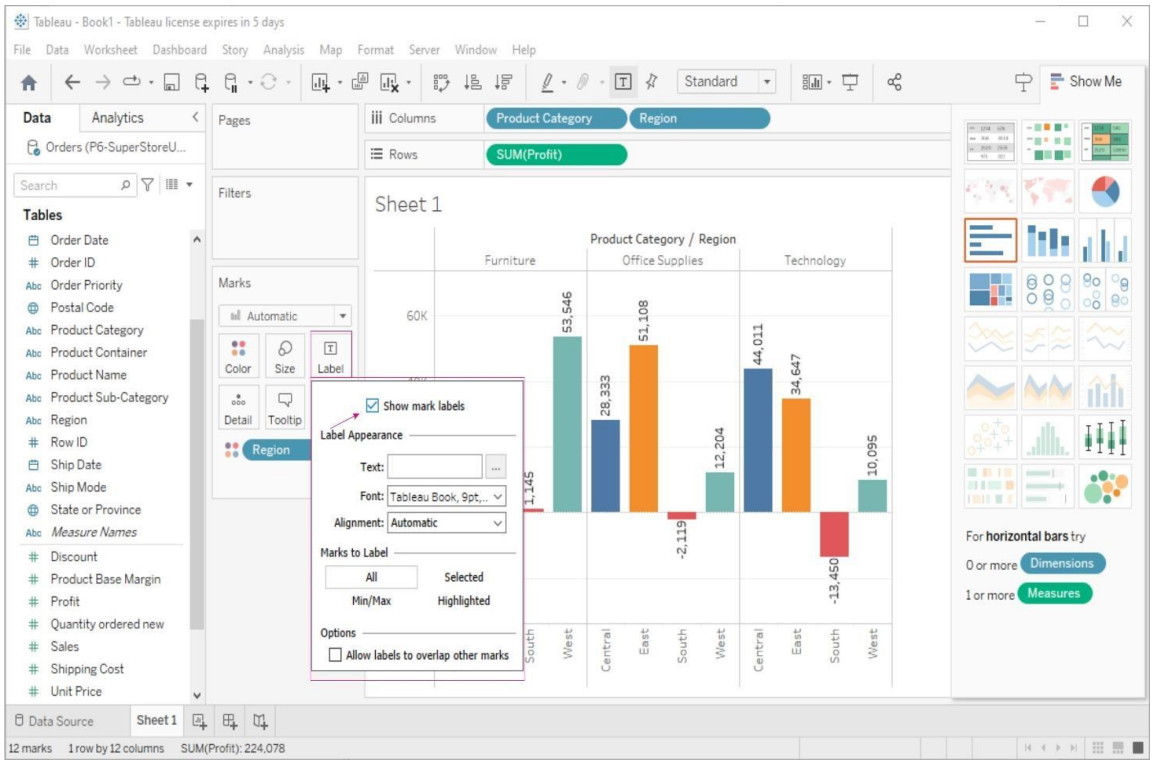


b. Bar Chart:

1. Drag and drop a **categorical field** (e.g., product category, region) to the Columns shelf and a **numeric field** to the Rows shelf.



2. Then **Tableau** will create a bar chart. You can adjust the orientation and formatting as needed. To display Labels on the bars click on Lables and select "**Show mark lables**"



6. Conclusion:

We have learnt about basic visualization techniques like bar chart, pie chart, line chart etc.

7. Viva Questions:

i. What are the basic visualization techniques?

ii. What visual elements are suitable for numerical and categorical dat??

References:

1. https://www.tableau.com/products/public
2. https://www.microsoft.com/en-us/power-platform/products/power-bi/desktop